

Notes: Mitchell & Mulherin (JFE, 1996)

The Impact of Industry Shocks on Takeover and Restructuring Activity

Contents

1	Big picture	2
2	Data and measurement	2
2.1	Motivation: takeover waves and industry shocks	2
2.2	Primary Value Line sample	3
2.3	Takeover and restructuring categories	3
2.4	CRSP robustness sample	3
2.5	Broad industry shock measures	4
2.6	Specific industry shock measures	4
3	Empirical specifications and tests	4
3.1	Cross-industry variation in takeover rates	4
3.2	Timing concentration within industries	5
3.3	Broad-shock regressions	5
3.4	Specific-shock regressions	5
4	Identification and interpretation	5
4.1	What identifies the industry-shock story	5
4.2	Why both rates and timing matter	6
4.3	How the paper differs from hostile-takeover explanations	6
4.4	Limits of the empirical design	6
5	Tables and figures	7
5.1	Table 1: Takeover activity during 1993–1994	7
5.2	Table 2: Frequency of takeover and restructuring activity in 1982–1989	7
5.3	Table 3 and Table 4: Industry variation in takeover rates	8
5.4	Table 5 and Table 6: Industry timing of takeover and restructuring activity	8
5.5	Table 7: Broad industry shocks	9
5.6	Table 8: Sources of change in industry structure	10
5.7	Table 9: Specific industry shocks	11
5.8	Appendix tables: robustness samples and shock measures	11
6	Short summary	15
7	Conclusion	16
7.1	Core conclusion	16
7.2	Economic intuition	16
7.3	Takeaway	16

1 Big picture

- **Question.** Why did takeover and restructuring activity in the 1980s concentrate in some industries rather than others? More specifically, were 1980s takeovers driven mainly by firm-specific discipline and opportunism, or by broader industry-level shocks?
- **Core hypothesis.** Corporate takeovers, mergers, leveraged buyouts, hostile bids, and asset restructurings are often low-cost mechanisms for reallocating assets when an industry experiences a fundamental shock.
- **Main empirical design.** The authors study 1,064 firms listed in the Value Line Investment Survey at year-end 1981 and track their takeover and restructuring activity during 1982–1989.
- **Industry-level perspective.** Instead of explaining individual takeover targets one-by-one, the paper asks whether takeover activity clusters by industry and whether high-activity industries are exposed to shocks such as deregulation, energy-price changes, foreign competition, or technological change.
- **First main fact.** Takeover and restructuring activity varies sharply across industries. Some industries have three-quarters or more of firms involved in some takeover or restructuring event; other industries have one-third or less.
- **Second main fact.** Activity is not only uneven across industries; it is also clustered over time within industries. In many industries, a large fraction of events occur in adjacent years.
- **Third main fact.** Industries with large sales and employment shocks in 1977–1981 have more takeover and restructuring activity in 1982–1989.
- **Specific shock evidence.** Deregulation is positively related to takeover activity, especially for actual takeovers. R&D intensity has a negative relation, consistent with industries with stronger internal adjustment/innovation being less likely to restructure through takeovers.
- **Important interpretation.** The 1980s takeover wave was not simply random, purely financial, or uniformly driven by hostile raiders. A large portion of activity appears to be an industry-level response to fundamental changes.
- **Broader implication.** Takeover waves should be studied as responses to broad economic shocks. Industry shocks can also explain spillover effects from takeover announcements, post-takeover performance patterns, and the timing of waves.

2 Data and measurement

2.1 Motivation: takeover waves and industry shocks

- The 1980s are often described as the fourth major U.S. takeover wave.
- Earlier descriptions of takeover waves tended to give each wave a label: monopoly mergers in the 1890s, oligopoly mergers in the 1920s, conglomerate mergers in the 1960s, and hostile bust-up takeovers in the 1980s.
- Mitchell and Mulherin argue that this description is incomplete because it ignores industry-specific shocks that make asset redeployment valuable.
- Examples of shocks include deregulation, changes in input prices, import competition, technological change, and financial innovation.

Interpretation: The key conceptual shift is from a firm-by-firm takeover story to an industry equilibrium story. If industries face shocks, then many firms in the same industry may need restructuring at roughly the same time.

2.2 Primary Value Line sample

- The main sample consists of 1,064 firms listed in the Value Line Investment Survey at year-end 1981.
- These firms are classified into 51 Value Line industries.
- Takeover and restructuring activity is tracked from 1982 through 1989.
- Activity is identified by searching financial periodicals for takeover and restructuring events.
- Market values are measured at year-end 1981 using the sum of market equity plus book debt.

Interpretation: Starting with firms in 1981 and following them through the 1980s makes the design forward-looking: the authors measure industry characteristics before the takeover wave and then ask whether subsequent activity follows those shocks.

2.3 Takeover and restructuring categories

The paper groups events into three broad categories:

1. **Friendly takeover targets.** These include mergers and leveraged buyouts not initially opposed by target management.
2. **Hostile takeover targets.** These include bids initially opposed by management, whether or not the hostile bid succeeds.
3. **Defensive asset restructurings.** These include major asset restructurings or financial recapitalizations induced by takeover pressure, even without a successful takeover.

The broad activity measure is

$$Activity_i = \mathbf{1}\{\text{firm } i \text{ experiences a takeover bid or major restructuring during 1982–1989}\}. \quad (1)$$

At the industry level, the equally weighted rate is

$$Rate_j = \frac{1}{N_j} \sum_{i \in j} Activity_i, \quad (2)$$

where N_j is the number of sample firms in industry j .

The value-weighted rate is

$$Rate_j^{vw} = \frac{\sum_{i \in j} MV_{i,1981} Activity_i}{\sum_{i \in j} MV_{i,1981}}, \quad (3)$$

where $MV_{i,1981}$ is firm i 's year-end 1981 value.

Interpretation: The equal-weighted rate asks how common takeovers are among firms in an industry. The value-weighted rate asks how much of the industry's economic value is exposed to takeover or restructuring.

2.4 CRSP robustness sample

- To verify the actual-takeover results outside the Value Line universe, the authors also analyze 3,660 CRSP-listed firms across 46 two-digit SIC industries.
- The CRSP sample focuses on actual takeover occurrences during 1982–1990.
- This broader sample is used as a robustness check for industry variation in actual takeovers.

Interpretation: The Value Line sample is rich because it identifies multiple event categories. The CRSP sample is broader and shows that the main industry clustering pattern is not an artifact of Value Line coverage.

2.5 Broad industry shock measures

The paper constructs pre-1982 shock measures using 1977–1981 Compustat data:

$$SalesGrowth_j = \ln \left(\frac{Sales_{j,1981}}{Sales_{j,1977}} \right), \quad (4)$$

$$SalesShock_j = |SalesGrowth_j - \overline{SalesGrowth}|. \quad (5)$$

Analogously,

$$EmploymentGrowth_j = \ln \left(\frac{Employment_{j,1981}}{Employment_{j,1977}} \right), \quad (6)$$

$$EmploymentShock_j = |EmploymentGrowth_j - \overline{EmploymentGrowth}|. \quad (7)$$

Interpretation: A shock is defined as a large deviation from the average industry trend, regardless of whether the deviation is positive or negative. The concept is that either rapid expansion or rapid contraction can force industry restructuring.

2.6 Specific industry shock measures

The paper studies several concrete shocks:

- **Deregulation:** indicator for industries undergoing major federal deregulation, including air transport, broadcasting/entertainment, natural gas, and trucking/transport leasing.
- **Energy dependence:** indicator for industries whose energy inputs are large or that are directly tied to energy markets.
- **Import vulnerability:** change in import penetration from 1977–1981 to 1982–1989.
- **R&D intensity:** research and development expense scaled by sales, used as a proxy for debt-financing ability or technological adjustment.

Interpretation: These specific shocks help move beyond the broad sales/employment shock variables and identify concrete forces behind restructuring.

3 Empirical specifications and tests

3.1 Cross-industry variation in takeover rates

The first test asks whether takeover activity is evenly distributed across industries. Let O_j be the observed number of events in industry j and E_j be the expected number under equal event rates. A Pearson chi-square statistic is

$$\chi^2 = \sum_j \frac{(O_j - E_j)^2}{E_j}. \quad (8)$$

The paper also uses likelihood-ratio tests and exact tests because expected cell counts are sometimes small.

Interpretation: Rejecting equal event rates means that takeover activity is not randomly distributed across industries. This is the first empirical requirement for an industry-shock explanation.

3.2 Timing concentration within industries

The second test asks whether takeover activity in a given industry is clustered in time. The paper uses two related tests:

- chi-square tests for equality of variances in timing across industries;
- ANOVA F -tests for equality of average event dates across industries.

For each industry, the authors also compute the maximum fraction of an industry's events occurring in adjacent two-year windows.

Interpretation: If all industries are merely participating in a common 1980s takeover wave, timing should be similar across industries. If industry shocks matter, event timing should cluster differently across industries.

3.3 Broad-shock regressions

The paper estimates regressions of the form

$$TakeoverRate_j = \alpha + \beta_1 Shock_j + \beta_2 Growth_j + \varepsilon_j. \quad (9)$$

It runs this specification using sales shocks and employment shocks. Dependent variables include:

- total takeover and restructuring activity;
- takeover attempts;
- actual takeovers.

Interpretation: The coefficient on $Shock_j$ tests whether industries that experienced larger pre-sample disturbances have more takeover activity during the 1980s.

3.4 Specific-shock regressions

The paper estimates regressions of the form

$$TakeoverRate_j = \alpha + \beta_1 Deregulation_j + \beta_2 Energy_j + \beta_3 ImportVulnerability_j + \beta_4 R\&D/Sales_j + \varepsilon_j. \quad (10)$$

Again, the dependent variables are total activity, takeover attempts, and actual takeovers.

Interpretation: These regressions ask which specific shocks account for cross-industry takeover activity after considering several channels jointly.

4 Identification and interpretation

4.1 What identifies the industry-shock story

- The main variation is across industries.
- Industry shocks are measured before or at the start of the takeover wave.
- Takeover and restructuring activity is measured after the shock window.
- The analysis compares industries with high versus low shock exposure.

Interpretation: This is not a causal natural experiment. It is a structured cross-industry test of whether takeover waves align with preexisting industry disturbances.

4.2 Why both rates and timing matter

- High rates alone could reflect permanent differences in industry governance or asset structure.
- Timing clustering alone could reflect macro takeover waves.
- The combination of high industry-specific rates, industry-specific timing, and relation to industry shocks is more persuasive.

Interpretation: The paper’s evidence is cumulative: rate variation, time clustering, and shock regressions all point toward the same industry-shock mechanism.

4.3 How the paper differs from hostile-takeover explanations

- Hostile takeovers and LBOs were prominent in the 1980s, but the paper argues that they are only part of a broader restructuring process.
- Friendly takeovers, hostile bids, actual acquisitions, recapitalizations, and asset restructurings all can be responses to industry shocks.
- The industry-level approach therefore broadens the analysis beyond corporate-control discipline at individual firms.

Interpretation: The corporate-control market is treated as an asset-reallocation mechanism, not only as a governance mechanism.

4.4 Limits of the empirical design

- Industry shocks are imperfect proxies; some shocks are qualitative or hard to measure.
- The regression sample has only 51 industries, so statistical power is limited.
- Takeover responses to shocks may differ across industries and may not occur immediately.
- The paper does not model the exact timing of each industry’s response to each shock.
- Results show strong association and consistency with the industry-shock mechanism, but they do not establish a fully causal structural model.

Interpretation: The paper’s strength is broad evidence on a historical wave; its limitation is that industry shocks are measured in reduced form.

5 Tables and figures

5.1 Table 1: Takeover activity during 1993–1994

Read:

- Seven industries account for 51.5% of takeover value during 1993–1994 but only 14.0% of total equity value.
- Banking and finance is the largest category, with 12.8% of takeover value and only 2.5% of equity value.
- Broadcasting, communications, leisure/entertainment, insurance, health services, and wholesale/distribution also account for disproportionate activity.

Economic message: The 1990s evidence motivates the paper: takeover waves are not uniform economy-wide events. They cluster in industries facing deregulation, technology, or other shocks.

Table 1
Takeover activity during 1993–94

This table lists seven industries that account for more than half of the takeover activity in the 1993–94 period for the 50 industries tracked by Mergerstat Review. Takeover value is in \$ millions. % of total takeover value is the value of takeovers in a particular Mergerstat industry divided by the total value of takeovers in the Mergerstat sample. % of total equity value is the value of the firms in a Mergerstat industry divided by the total equity value at year-end 1992 for all publicly traded firms on CRSP. The matching between CRSP and Mergerstat is done via SIC codes. (Mergerstat Review provides the SIC codes associated with each Mergerstat industry; CRSP provides the primary SIC code for each listed firm.)

Industry	Takeover value	% of total takeover value	% of total equity value
Banking & finance	\$ 51,637	12.8%	2.5%
Broadcasting	\$ 43,129	10.7	0.8
Communications	\$ 27,661	6.9	4.3
Leisure & entertainment	\$ 25,339	6.3	1.5
Insurance	\$ 22,017	5.5	2.4
Health services	\$ 21,896	5.4	1.0
Wholesale & distribution	\$ 15,704	3.9	1.5
Sum for 7 industries	\$207,383	51.5	14.0
Total takeover value	\$403,071		

Table 2
Frequency of takeover and restructuring activity during 1982–89

This table presents the incidence of takeover and restructuring activity for a sample of 1,064 firms listed in the Value Line Investment Survey at year-end 1981. Takeover and restructuring activity in the 1982–89 period is determined by searching financial periodicals. Takeovers include tender offers, mergers, and leveraged buyouts. Friendly takeovers are those not initially opposed by target management. Hostile takeovers are those initially opposed by target management. Defensive asset restructurings include any major asset restructuring or recapitalization induced by implicit takeover pressure, such as a large block purchase by a corporate raider or the growing occurrence of takeover activity in the firm's industry. The value-weighted % is the total value (market equity plus book debt) of the firms in the respective category divided the total value of the full sample, for which all values are recorded at year-end 1981.

Classification	Number of firms	Equally weighted %	Value-weighted %
Friendly takeover target	286	26.9%	13.6%
Successful	268	25.2	13.0
Unsuccessful	18	1.7	0.6
Hostile takeover target	243	22.8%	23.4%
Successful	85	8.0	7.3
Unsuccessful	35	3.3	2.2
Unsuccessful followed by friendly takeover	87	8.1	7.0
Unsuccessful followed by restructuring	36	3.4	6.9
Defensive asset restructuring	78	7.3%	11.8%
Asset restructuring	64	6.0	9.0
Financial recapitalization	14	1.3	2.8
Remainder of the sample	457	43.0%	51.2%
Total sample	1,064	100.0%	100.0%

In 15 successful friendly takeovers, the bidder gained control without an actual takeover. In four unsuccessful friendly takeovers, the firms subsequently engaged in major asset restructurings.

5.2 Table 2: Frequency of takeover and restructuring activity in 1982–1989

Read:

- In the 1,064-firm Value Line sample, 57.0% of firms experience some takeover or restructuring activity during 1982–1989.
- 26.9% are friendly takeover targets, 22.8% are hostile takeover targets, and 7.3% undertake defensive asset restructurings.
- Actual control changes occur for about 41% of firms in the sample.
- Value-weighted activity is similar: about 48.8% of firm value is associated with takeover or restructuring activity.

Economic message: The 1980s wave is broad, but the paper's key question is whether this broad activity is concentrated by industry and linked to shocks.

5.3 Table 3 and Table 4: Industry variation in takeover rates

Read:

- Table 3 shows large cross-industry variation in takeover and restructuring activity.
- Industries such as Entertainment, Drugstores, Petroleum Producing, Broadcasting, Textiles, Tire and Rubber, and Integrated Steel have very high activity rates.
- Other industries, such as Paper and Paper Products, Specialty Chemicals, Metal Fabricating, and Leather, have much lower rates.
- Table 4 formally rejects the null of no cross-industry variation in total activity, takeover attempts, and actual takeovers.
- Exact-test p-values confirm that the rejection is not an artifact of small expected cell counts.

Economic message: Takeover activity is not randomly distributed across industries. This supports the idea that industry-level forces shape the market for corporate control.

Table 3
Cross-sectional variation in takeover and restructuring activity by Value Line industry during 1982-89

This table presents the takeover and restructuring activity during 1982-89 across 51 industries encompassing 1,064 firms listed in the Value Line Investment Survey at year-end 1981. Takeover and restructuring activity in the 1982-89 period is determined by searching financial periodicals. Industry value is the sum of the total value (book debt plus market equity, in \$ millions) of the firms in the industry as of year-end 1981. Total activity is the fraction of firms in an industry that experience a takeover bid or a major restructuring in the 1982-89 period. Takeover attempts is the fraction of firms that receive a takeover bid. Friendly bids and hostile bids respectively segregate the fraction of takeover bids into friendly and hostile categories. Actual takeovers reports the fraction of firms that actually realize a change in control. The "Equal" columns show the % of firms in an industry that experience activity. The "Value" columns show the proportion of total value in an industry, based on year-end 1981 values, that experiences activity in the 1982-89 period. The data are presented in descending order of the fraction of firms experiencing takeover or restructuring activity.

Industry	Number of firms	Industry value	Total activity		Takeover attempts		Friendly bids		Hostile bids		Actual takeovers	
			Equal	Value	Equal	Value	Equal	Value	Equal	Value	Equal	Value
Entertainment	17	11,204	82%	88%	71%	83%	47%	22%	24%	61%	59%	61%
Drugstores	10	3,525	80	79	70	67	60	51	10	16	70	67
Petroleum producing	19	17,972	79	91	63	71	26	14	37	57	47	63
Broadcasting	13	8,200	77	83	77	83	38	42	38	41	69	64
Textiles	17	4,248	77	82	76	82	41	22	35	60	71	81
Tire & rubber	12	7,464	75	90	67	75	17	4	50	71	50	31
Integrated steel	12	17,441	75	81	33	44	17	12	17	32	17	9
Air transport	23	18,462	74	70	70	70	39	42	30	28	48	41
Toiletries & cosmetics	15	7,011	73	97	60	91	27	7	33	84	47	47
Metals & mining	11	13,862	73	57	64	43	0	0	64	43	18	16
Natural gas	21	41,647	71	76	67	53	29	17	38	36	57	45
Precision instruments	21	17,811	71	28	62	14	48	8	14	6	43	8
Retail stores, special lines	17	6,910	71	34	65	20	35	15	29	15	47	26
Grocery stores	20	10,662	70	72	70	72	35	19	35	53	65	60
Truck & transport leasing	20	7,118	70	58	45	42	15	5	30	37	45	42
Integrated petroleum	36	214,134	69	60	56	41	17	8	39	34	39	28
Hotels & gaming	12	8,000	67	81	42	38	8	6	33	32	17	10
Cool, uranium, geothermal	9	4,620	67	56	44	36	22	4	22	32	22	4
Construction machinery	17	11,740	65	27	65	27	24	9	41	17	53	20

202

M.L. Mitchell, J.H. Mulherin / Journal of Financial Economics 41 (1996) 193-229

Multiform	40	38,755	63	79	50	40	28	20	23	20	45	38
Building	58	23,195	62	65	52	44	29	19	22	26	40	29
Replacement auto parts	15	5,710	60	48	47	37	27	12	20	25	33	33
Home appliances	10	2,740	60	44	60	44	50	26	10	22	60	44
Original equip. auto parts	10	3,807	60	34	50	23	40	22	10	1	50	23
Packaging & containers	22	9,206	59	70	50	67	23	20	27	47	45	66
Publishing	22	10,778	59	48	59	48	32	16	27	32	50	24
Oilfield services	27	37,193	59	52	41	37	30	32	11	5	30	30
Apparel	26	3,987	58	74	54	68	35	52	19	16	46	63
Fast food & restaurants	16	6,729	56	31	50	28	31	19	19	9	50	28
Medical services	9	9,916	56	59	56	59	44	47	11	12	56	59
Food processing	54	40,722	56	57	52	51	28	25	24	26	48	50
Machinery	38	16,757	55	48	50	38	37	27	13	12	47	31
Retail stores	38	35,333	55	30	50	28	29	7	22	21	39	19
Industrial services	19	6,713	53	25	53	25	47	23	5	2	47	24
Computer data processing	31	63,916	52	14	45	11	19	2	26	9	45	11
Shoes	12	4,515	50	44	33	35	17	12	17	23	17	12
Mig. bus. & rec. vehicles	12	890	50	37	42	36	42	36	0	0	42	36
General steel	10	1,725	50	34	40	28	20	14	20	14	40	28
Aerospace & defense	28	24,383	50	34	46	25	25	8	21	17	39	18
Basic chemicals	13	45,656	46	26	38	25	15	7	23	18	23	7
Electrical equipment	34	31,430	44	22	38	8	26	6	12	2	38	8
Diversified chemicals	14	21,212	43	41	21	12	7	7	14	5	21	12
Toys & school supplies	10	1,410	40	29	40	29	30	23	10	6	40	29
Health care & hosp. supply	23	19,865	39	16	35	14	13	3	22	12	35	14
Drugs	13	28,814	38	31	38	31	31	30	8	1	38	31
Office equipment & supplies	13	7,243	38	19	38	19	31	13	8	6	38	19
Machine tool	16	5,378	38	21	38	21	6	1	31	20	31	20
Electronics	41	26,688	37	25	37	25	22	20	15	5	29	23
Paper & paper products	27	32,245	35	24	31	24	7	4	26	20	20	21
Specialty chemicals	22	8,058	27	14	27	14	9	5	18	9	14	8
Metal fabricating	19	4,463	26	21	26	21	16	15	11	6	26	21
All firms	1,064	1,013,512	57%	49%	50%	37%	27%	14%	23%	23%	41%	28%

M.L. Mitchell, J.H. Mulherin / Journal of Financial Economics 41 (1996) 193-229 203

Table 4
Tests comparing takeover and restructuring frequency across industries

This table reports the probability values from tests assessing the null of no variation in takeover and restructuring activity across 51 Value Line industries during the 1982-89 period, and actual takeovers across 46 CRSP SIC industries during the 1982-90 period. Pearson refers to the Pearson chi-square statistic that pertains to the difference between the actual and the expected proportions of corporate control activity across industries: $O_{ij} = \sum_j \sum_i (Actual_{ij} - Expected_{ij})^2 / Expected_{ij}$. Likelihood ratio refers to the likelihood ratio chi-square statistic that pertains to the ratio of the actual and expected proportions across industries: $G_2 = 2 \sum_j \sum_i Actual_{ij} \ln(Actual_{ij}/Expected_{ij})$. Exact test is based on the simulation of probability distributions using algorithms developed by Mehta and Patel (1983). Degrees of freedom are 50 for the three Value Line categories and 45 for the CRSP sample. A description of the Value Line takeover and restructuring categories is provided in Table 3. The CRSP sample is described in Table 10 in the Appendix.

Category	Probability value		
	Pearson	Likelihood ratio	Exact test
Total activity	0.003	0.002	0.002
Takeover attempts	0.081	0.060	0.064
Actual takeovers	0.069	0.042	0.040
Actual takeovers, CRSP sample	0.001	0.001	0.001

5.4 Table 5 and Table 6: Industry timing of takeover and restructuring activity

Read:

- Table 5 reports event counts by industry and year from 1982 through 1989.
- Many high-activity industries have a large share of events in adjacent two-year periods.
- Original Equipment Auto Parts has 100% of events in a two-year window; Home Appliances has 83%; Drugstores has 75%.

- Table 6 rejects equality of timing dispersion and equality of mean event dates across industries.
- The results hold for total activity, takeover attempts, actual takeovers, and actual takeovers in the CRSP sample.

Economic message: Takeover activity is clustered not only in which industries are affected, but also in when industries are affected. This is what one expects if industry shocks trigger restructuring waves.

Table 5
Takeover and restructuring activity by year for Value Line industries

This table reports the number of takeovers and restructurings by industry by year for a sample of 1,064 firms from 51 industries listed on Value Line as of fourth quarter 1981. A takeover or restructuring is assigned to the year in which the initial announcement of the event occurs. Maximum cluster % in 2 years is the greatest fraction of takeovers and restructurings occurring in an industry in an adjacent two-year period.

Industry	Number of takeovers and restructurings per year								Maximum cluster % in 2 years
	82	83	84	85	86	87	88	89	
Orig. equip. auto parts	0	0	0	2	4	0	0	0	100%
Home appliances	0	0	0	1	4	0	1	0	83
Drugstores	0	0	4	2	1	0	1	0	75
Paper & paper products	0	0	1	3	3	0	1	1	67
Basic chemicals	0	0	0	2	2	0	1	1	67
Specialty chemicals	0	0	0	0	1	1	2	2	67
Air transport	0	2	1	1	6	5	1	1	65
Grocery stores	1	0	2	0	2	5	4	0	64
Construction machinery	1	0	1	0	3	4	1	1	64
Drugs	0	0	0	1	0	1	0	3	60
Broadcasting	0	2	0	4	2	1	0	1	60
Metal fabricating	0	1	0	0	1	0	3	0	60
Replacement auto parts	2	0	0	0	1	1	2	3	56
Tire & rubber	0	1	1	1	1	3	2	0	56
Petroleum producing	2	5	3	3	1	1	0	0	53
Electronics	2	0	0	1	2	2	4	4	53
Oilfield services	1	1	7	1	4	0	0	2	50
Entertainment	4	3	2	2	1	0	0	2	50
Mfg. hous. & rec. vehicle	3	0	0	0	0	1	1	1	50
Aerospace & defense	4	0	1	3	4	1	1	0	50
Metals & mining	1	0	0	2	0	2	2	1	50
Coal, uranium, geothermal	1	1	1	2	0	0	0	1	50
Shoes	0	0	0	1	1	2	1	1	50
Toys & school supplies	1	0	2	0	0	0	1	0	50
Diversified chemicals	0	1	0	0	2	1	2	0	50
Hotels & gaming	0	0	0	2	2	2	1	1	50
Multiform	2	5	7	3	1	4	2	1	48
Precision instruments	1	2	0	0	4	3	3	2	47
Electrical equipment	0	0	2	2	4	3	3	1	47
Fast food & restaurants	1	0	2	0	1	1	2	2	44
Integrated steel	1	3	1	1	3	0	0	0	44
Health care & hosp. supply	3	0	1	3	0	0	2	0	44
Computer data processing	0	3	2	1	4	3	1	2	44
Retail stores	4	1	8	1	4	1	2	0	43
Retail stores, special line	2	0	1	0	4	1	1	3	42
Medical services	0	1	1	0	1	1	1	0	40
General steel	2	0	0	1	1	0	1	0	40
Apparel	1	2	3	3	2	1	2	1	40
Office equip. & supplies	1	1	1	1	1	0	0	0	40

Table 5 (continued)

Industry	Number of takeovers and restructurings per year								Maximum cluster % in 2 years
	82	83	84	85	86	87	88	89	
Publishing	2	1	3	0	1	1	3	2	38
Packaging & containers	0	1	2	3	2	2	1	2	38
Textiles	3	2	2	1	1	2	2	0	38
Truck & transport leasing	1	2	2	2	3	2	2	1	36
Toiletries & cosmetics	0	1	1	2	2	1	1	3	36
Natural gas	2	3	1	3	2	1	1	2	33
Building	3	5	7	3	6	1	6	5	33
Food processing	4	4	3	6	2	2	8	1	33
Machinery	4	3	2	4	3	2	1	2	33
Machine tool	1	1	0	1	1	0	0	2	33
Integrated petroleum	3	4	5	3	4	1	5	0	32
Industrial services	3	0	3	0	3	0	1	0	30
Annual total	67	62	86	78	108	65	83	58	
% of all takeovers & restructurings	11.0	10.2	14.2	12.9	17.8	10.7	13.7	9.6	

of a two-year window is somewhat arbitrary, alternate groupings (e.g., a three-year period) yield similar conclusions.

Table 6
Variation in the timing of takeover and restructuring activity

This table reports tests of the null hypothesis of no variation in the timing of takeover and restructuring activity for three takeover categories across 51 Value Line industries during 1982–89, and for actual takeovers across 46 CRSP SIC industries during 1982–90. Panel A reports results from chi-square tests that compare the variance of the timing of takeover activity for an entire sample with that within control groups (i.e., industries). Panel B reports results from ANOVA *F*-tests for the equality of mean takeover dates across industries. Degrees of freedom are 50 for the Value Line categories and 45 for the CRSP sample.

Panel A: Chi-square tests for the equality of variance

Category	Chi-square statistic	Probability value
Total activity	98.49	0.0001
Takeover attempts	103.20	0.0001
Actual takeovers	87.44	0.0008
Actual takeovers, CRSP sample	261.00	0.0001

Panel B: ANOVA *F*-tests for the equality of means

Category	<i>F</i> -statistic	Probability value
Total activity	2.02	0.0001
Takeover attempts	1.77	0.0015
Actual takeovers	1.53	0.0155
Actual takeovers, CRSP sample	1.67	0.0040

5.5 Table 7: Broad industry shocks

Read:

- Table 7 regresses takeover activity on sales shocks and employment shocks measured before the takeover wave.
- In Panel A, the sales shock coefficient is positive for total activity, takeover attempts, and actual takeovers.
- The coefficient is about 0.46–0.47, with t-statistics around 2.10 to 2.74.
- Sales growth itself is not significant, which means it is the magnitude of the deviation, not high or low growth per se, that matters.
- Panel B finds similar positive effects for employment shocks.

Economic message: Both positive and negative industry disturbances lead to restructuring. The market for corporate control responds to change, not simply to decline or boom conditions.

Table 7
Regressions of takeover activity on broad industry shocks

This table reports regressions of industry measures of takeover activity on measures of industry shocks based on sales and employment data from the 1977–81 period. Panel A regresses takeover activity on industry sales shocks, and also industry sales growth, a proxy for industry performance. Panel B regresses takeover activity on industry employment shocks, and also industry employment growth, a proxy for industry performance. The definitions of the three takeover and restructuring categories are provided in Table 3. Industry sales and employment shocks and growth are defined in Table 11 in the Appendix. Each regression uses 51 observations. *t*-statistics are in parentheses.

<i>Panel A: Takeover activity, sales shocks, and sales growth</i>			
	Total activity	Takeover attempts	Actual takeovers
Intercept	44.08 (8.92)	36.39 (8.34)	27.02 (7.07)
Sales shock	0.46 (2.10)	0.47 (2.38)	0.47 (2.74)
Sales growth	− 0.04 (− 0.26)	− 0.11 (− 0.72)	− 0.11 (− 0.82)
Adjusted R^2	0.05	0.07	0.10

<i>Panel B: Takeover activity, employment shocks, and employment growth</i>			
	Total activity	Takeover attempts	Actual takeovers
Intercept	46.64 (10.34)	36.81 (9.36)	26.95 (7.74)
Employment shock	0.45 (1.77)	0.53 (2.39)	0.51 (2.60)
Employment growth	− 0.22 (− 1.16)	− 0.21 (− 1.26)	− 0.14 (− 0.93)
Adjusted R^2	0.03	0.07	0.09

5.6 Table 8: Sources of change in industry structure

Read:

- Table 8 documents qualitative sources of structural change in high-takeover industries.
- Air transport, broadcasting/entertainment, natural gas, and trucking are linked to deregulation.
- Energy-related sectors are linked to price volatility following the 1973 OPEC embargo and 1978–1979 Iranian oil-export cutoff.
- Apparel/textiles and metals/mining are linked to import competition or foreign competition.
- Health-related industries are linked to Medicare reimbursement changes.

Economic message: The qualitative table supports the statistical evidence: high takeover industries are often industries undergoing clear economic transformations.

Table 8

Sources of change in industry structure

This table reports reasons offered for the changing structure of high-takeover industries during the 1980s. Reference sources for this table include the Value Line Investment Survey, S&P Industry Outlooks, the U.S. Industrial Outlook, the *Wall Street Journal*, and other analyst reports.

Industry	Source of change
Air transport	Airline Deregulation Act of 1978 phased out the government control of airline routes and pricing during the early 1980s. The deregulation facilitated entry, the development of the hub-and-spoke system, and the movement to nationwide route systems.
Apparel, Textiles	A significant increase of imports, induced in part by a shift to an antiprotectionist foreign trade policy, prompted consolidation as a means of streamlining operations to compete effectively.
Broadcasting, Entertainment	FCC policies in the early 1980s eliminated programming and ownership regulations that had limited the number of radio and television stations held by one corporation, and had impeded the sale of radio and television stations. Cable Communications Policy Act of 1984 phased out rate restrictions on cable TV.
Coal, uranium, geothermal	Changes in petroleum prices first induced expansion, and later contraction, in these energy sectors.
Drugstores	Moderate growth prospects and new competition from grocery stores and discounters (such as Walmart) induced consolidation among industry members.
Food processing	Excess capacity in mature industry due to low population growth rate led to consolidation.
Grocery stores	Declining industry growth, relaxed antitrust environment, high tangibility of assets, and ability to use leverage financing led to change in industry structure.
Hotels & gaming	Volatility in real estate and building prices.
Integrated petroleum, Oilfield services, Petroleum producing	1973 OPEC embargo and 1978-79 Iranian oil-export cutoff created price volatility that ultimately caused industry restructuring and consolidation.
Integrated steel	Secular decline in steel demand and subsidized foreign competition resulted in excess capacity and consolidation.
Medical services	The implementation of Medicare's Prospective Payment System in 1983 changed reimbursement from cost-based to a fixed payment per diagnosis. This inducement to curb costs prompted alterations in industry structure and promoted the growth of HMOs.
Metals & mining	Foreign competition, brought on in part by subsidization by foreign governments, resulted in excess capacity and subsequent industry consolidation.

Table 8 (continued)

Multiform	Conglomerates formed in the 1960s and 1970s restructured in the 1980s.
Natural gas	The Natural Gas Policy Act of 1978 phased in price deregulation and encouraged the combination of pipelines to enable regional and national gas transportation systems.
Packaging & containers	Foreign competition and the realignment from metal and glass led to industry restructuring.
Tire & rubber	Longer-lasting radial tires, rising imports, and the reduced demand for automobiles following energy price increases created excess capacity and subsequent industry consolidation.
Truck & transport leasing	Trucking deregulation of 1980 allowed new entry, resulting in excess capacity that led to consolidation.
Toiletries & cosmetics	Demographic changes slowed industry growth and led to consolidation.

Table 8, first page

Table 8, continued

5.7 Table 9: Specific industry shocks

Read:

- Deregulation has a positive relation with takeover and restructuring activity.
- In total activity regressions, the deregulation coefficient is about 28.24 when entered alone and remains positive in the multivariate specification.
- R&D/sales is negatively related to takeover activity. In the multivariate specification, it is negative and statistically significant for total activity, takeover attempts, and actual takeovers.
- Energy dependence and import vulnerability do not explain much of the cross-industry variation once other variables are included.
- Actual takeover regressions in Panel C are very similar to the takeover-attempt regressions.

Economic message: Deregulation is a concrete driver of takeover activity. R&D-intensive industries may adjust internally rather than through takeover markets, or they may be harder to finance with takeover debt.

5.8 Appendix tables: robustness samples and shock measures

Read:

- Table 10 documents actual takeovers across 46 two-digit SIC industries in the CRSP sample.

Table 9

Regressions of takeover activity on specific industry shocks

This table reports regressions of industry measures of takeover activity on specific measures of industry shocks, including deregulation, energy dependence, foreign competition, and ability to debt-finance. Panel A reports regressions for total takeover and restructuring activity. Panel B reports regressions for takeover attempts. Panel C reports regressions for actual takeovers. Deregulation is a dummy variable for five industries experiencing major federal deregulation: Air transport, Broadcasting, Entertainment, Natural gas, and Trucking & transport leasing. Energy dependence is a dummy variable for energy-related industries and any industry in which energy accounts for 10% or more of input costs as described in Table 12 in the Appendix. Import vulnerability is a dummy variable equal to one for the top third of the industries described in Table 13 that experience the greatest increase in import penetration in the 1980s. R&D/sales (for 1981) proxies for debt-financing ability and is taken from Compustat. Each regression uses 51 observations. *t*-statistics are in parentheses.

Panel A: Total takeover and restructuring activity

	(1)	(2)	(3)	(4)	(5)
Intercept	46.76 (13.82)	46.89 (13.21)	51.18 (12.24)	59.42 (15.81)	54.92 (11.32)
Deregulation	28.24 (2.61)	—	—	—	17.90 (1.61)
Energy dependence	—	19.26 (2.01)	—	—	4.82 (0.49)
Import vulnerability	—	—	− 4.94 (− 0.68)	—	2.60 (0.40)
R&D/sales	—	—	—	− 6.77 (− 4.22)	− 5.94 (− 3.55)
Adjusted R^2	0.10	0.06	− 0.01	0.25	0.27

Panel B: Takeover attempts

	(1)	(2)	(3)	(4)	(5)
Intercept	37.96 (12.79)	39.25 (12.09)	42.71 (11.49)	48.96 (14.25)	46.58 (10.70)
Deregulation	28.24 (2.98)	—	—	—	23.59 (2.37)
Energy dependence	—	10.75 (1.23)	—	—	− 5.62 (− 0.64)
Import vulnerability	—	—	− 5.94 (− 0.92)	—	− 0.34 (− 0.06)
R&D/sales	—	—	—	− 5.64 (− 3.84)	− 4.98 (− 3.32)
Adjusted R^2	0.14	0.01	− 0.003	0.22	0.26

Table 9 (continued)

Panel C: Actual takeovers

	(1)	(2)	(3)	(4)	(5)
Intercept	29.33 (10.82)	30.66 (10.49)	32.59 (9.79)	37.40 (11.65)	35.82 (8.67)
Deregulation	21.37 (2.46)	—	—	—	19.59 (2.08)
Energy dependence	—	5.48 (0.70)	—	—	− 7.28 (− 0.88)
Import vulnerability	—	—	− 3.53 (− 0.61)	—	0.30 (0.06)
R&D/sales	—	—	—	− 4.10 (− 2.99)	− 3.72 (− 2.61)
Adjusted R^2	0.09	− 0.01	− 0.01	0.14	0.16

The analysis of actual takeovers in panel C of Table 9 is nearly identical to the

- Table 11 reports industry sales and employment growth and shock measures for 1977–1981.
- Table 12 reports energy dependence by industry based on input-output data.
- Table 13 reports changes in import penetration ratios.

Economic message: The appendix provides the raw industry-level measures underlying the regressions. This makes the shock definitions transparent and shows that the main conclusions are not only narrative.

Appendix

Table 10

Cross-sectional frequency of actual takeover occurrences during 1982–90 for CRSP-listed firms

This table reports the occurrence of actual takeovers across a sample of 46 two-digit SIC industries encompassing 3,660 firms taken from the CRSP tapes. Equal reports the fraction of takeover occurrences. Equity reports the fraction of the equity of an industry that was acquired.

Industry	SIC code	Number of firms	Number of takeovers	% takeovers	
				Equal	Equity
Food stores	54	53	32	60.4	56.2
Railroad transportation	40	15	9	60.0	52.8
Textile mill products	22	53	28	52.8	71.3
General merchandise	53	56	27	48.2	23.6
Miscellaneous retail	59	56	26	46.4	65.8
Stone, clay, glass, concrete	32	62	28	45.2	41.5
Hotels	7	35	15	42.9	44.9
Motor freight transportation	42	31	13	41.9	21.8
Non-building construction	16	17	7	41.2	27.1
Apparel	23	69	28	40.6	60.7
Furniture & home furnishings	57	15	6	40.0	6.4
Food products	20	122	48	39.3	29.6
Paper products	26	54	21	38.9	21.4
Transportation services	47	13	5	38.5	7.9
Transportation equipment	37	95	36	37.9	15.5
Personal services	72	16	6	37.5	12.3
Eating & drinking places	58	64	24	37.5	22.5
Building materials	52	16	6	37.5	52.4
Apparel & accessory stores	56	32	12	37.5	35.9
Air transportation	45	40	15	37.5	30.0
Primary metal	33	91	34	37.4	19.0
Rubber & plastic products	30	62	23	37.1	23.8
Health services	80	45	16	35.6	53.3
Motion pictures & video	78	30	10	33.3	56.3
Miscellaneous manufacturing	39	65	21	32.3	50.9
Medical & photographic equipment	38	214	69	32.2	16.5
Fabricated metal products	34	156	50	32.1	38.0
Building construction	15	25	8	32.0	16.3
Amusement & recreation services	79	22	7	31.8	49.5
Furniture & fixtures	25	33	10	30.3	17.6
Machinery, except electrical	35	328	96	29.3	20.5
Chemicals	28	182	51	28.0	19.2
Wholesale trade, durables	50	119	33	27.7	13.7
Communications	48	58	16	27.6	13.3
Printing & publishing	27	77	21	27.3	10.1
Business services	73	219	59	26.9	30.9

Table 10 (continued)

Industry	SIC code	Number of firms	Number of takeovers	% takeovers	
				Equal	Equity
Wholesale trade, nondurables	51	54	13	24.1	15.8
Electrical machinery	36	339	81	23.9	14.4
Lumber & wood products	24	44	10	22.7	17.3
Petroleum refining	29	45	10	22.2	28.7
Leather	31	24	5	20.8	3.7
Oil & gas extraction	13	389	82	20.6	35.3
Metal mining	10	67	10	14.9	18.4
Engineering & misc. services	89	34	5	14.7	44.2
Special-trades construction	17	14	2	14.3	39.0
Coal mining	12	10	1	10.0	6.8
All firms		3,660	1,133	31.0%	27.7%

Table 11

Industry sales and employment shocks, 1977–81

This table reports values for sales and employment growth and sales and employment shocks over the 1977–81 period for 51 Value Line industries. The data are sorted based on the sales shock variable. Sales growth is the real growth in sales between 1977 and 1981 for each industry, computed as $\ln(\text{Sales } 1981 / \text{Sales } 1977)$, where sales values are put in real terms using the GNP deflator. Sales shock is the absolute value of the difference between an industry's sales growth and the average industry sales growth of 18.12% for the full sample of 51 industries. Employment growth is the growth in the number of employees between 1977 and 1981 for each industry, computed as $\ln(\text{Employment } 1981 / \text{Employment } 1977)$. Employment shock is the absolute value of the difference between an industry's employment growth and the average industry employment growth of 10.40% for the full sample of 51 industries. Industry members are taken from firms listed on Value Line in the fourth quarter of 1981. Sales and employment data are taken from Compustat.

Industry	Sales growth	Sales shock	Employment growth	Employment shock
Medical services	100.70%	82.58%	99.03%	88.63%
Mfg. housing & rec. vehicles	−42.52	60.64	−29.26	39.66
Petroleum producing	64.54	46.42	−6.25	16.65
Tire & rubber	−15.84	33.96	−26.59	36.99
Natural gas	50.98	32.85	16.77	6.37
Textiles	−11.40	29.52	−16.51	26.92
Oilfield services	42.30	24.18	23.24	12.83
Integrated petroleum	41.47	23.35	18.15	7.74
Original equipment auto parts	−4.88	23.01	−3.74	14.14
Coal, uranium, geothermal	41.01	22.88	5.30	5.11

Table 10, first page

Table 10 continued and Table 11 start

Table 11 (continued)

Industry	Sales growth	Sales shock	Employment growth	Employment shock
Apparel	- 1.77	19.90	- 7.93	18.34
Home appliances	- 0.65	18.77	- 14.39	24.79
Replacement auto parts	1.61	16.52	8.06	2.35
Hotels & gaming	34.62	16.50	30.35	19.94
Paper & paper products	3.45	14.68	- 4.66	15.07
Packaging & containers	4.18	13.94	- 4.49	14.90
Construction machinery	4.45	13.67	- 1.66	12.06
Air transport	30.88	12.75	18.23	7.82
Toys & school supplies	30.55	12.43	15.54	5.13
Integrated steel	5.71	12.42	- 5.70	16.11
Industrial services	30.13	12.01	22.01	11.60
Publishing	29.37	11.24	16.50	6.09
Broadcasting	8.25	9.88	15.17	4.76
Metals & mining	8.33	9.80	- 0.41	10.82
Food processing	8.57	9.55	15.03	4.63
Grocery stores	9.18	8.94	17.60	7.20
Basic chemicals	26.79	8.67	12.17	1.76
Diversified chemicals	9.58	8.55	12.33	1.93
Entertainment	10.43	7.69	4.22	6.18
Retail stores	11.27	6.85	5.20	5.20
Trucking & transport leasing	12.38	5.74	7.04	3.36
Machine tool	23.71	5.58	12.83	2.42
Health care & hosp. supply	23.47	5.35	15.59	5.19
Machinery	13.45	4.68	5.49	4.92
Metal fabricating	22.40	4.28	12.69	2.28
Electrical equipment	13.59	4.17	7.43	2.97
Computer data processing	22.23	4.11	23.49	13.09
Retail stores, special lines	22.03	3.91	24.66	14.25
Fast food & restaurants	21.97	3.84	29.78	19.38
Multiform	15.30	2.82	9.68	0.72
Drugstores	20.75	2.62	25.71	15.31
Precision instruments	20.60	2.48	10.60	0.20
Specialty chemicals	15.95	2.17	10.56	0.16
Drugs	16.29	1.83	6.28	4.13
General steel	19.92	1.79	- 2.89	13.30
Aerospace & defense	19.64	1.52	12.61	2.21
Building	16.66	1.47	8.46	1.94
Electronics	19.53	1.41	20.82	10.42
Shoes	16.85	1.27	23.43	13.03
Toiletries & cosmetics	17.89	0.23	18.07	7.67
Office equipment & supplies	18.04	0.08	14.99	4.58
Average	18.12%	13.44%	10.40%	11.63%

Table 11 continued

Table 12

Energy dependence by industry

This table reports the dependence of an industry on energy inputs for the 51 Value Line industries. Energy fraction is the fraction of an industry's inputs comprised of energy inputs. Data are for 1982, and are taken from the 1982 *Benchmark Input-Output Accounts of the United States*, U.S. Department of Commerce, December 1991, and 'Benchmark Input-Output Accounts for the U.S. Economy, 1982', *Survey of Current Business*, Volume 71, July 1991. Energy inputs are defined as those coming from Coal (industry # 7), Crude petroleum (industry # 8), and Petroleum refining (industry # 31). The Value Line industries are matched with the industries used in the input-output tables via primary and secondary SIC codes.

Value Line industry	Input-output industry	Energy fraction
Integrated petroleum	Petroleum refining	73.92%
Natural gas	Gas production & distribution (utilities)	35.33
Air transport	Air transportation	25.26
Coal, uranium, geothermal	Coal	15.78
Trucking & transport leasing	Motor freight transportation & warehousing	10.07
Metals & mining	Iron & ferroalloy ores	7.08
Basic chemicals	Chemicals & selected chemical products	5.73
Diversified chemicals	Chemicals & selected chemical products	5.73
Specialty chemicals	Chemicals & selected chemical products	5.73
General steel	Primary iron & steel manufacturing	4.74
Integrated steel	Primary iron & steel manufacturing	4.74
Paper & paper products	Paper & allied products	3.74
Oilfield services	Crude petroleum & natural gas	3.63
Petroleum producing	Crude petroleum & natural gas	3.63
Building	New construction	3.41
Drugstores	Retail trade	2.26
Grocery stores	Retail trade	2.26
Retail stores, special lines	Retail trade	2.26
Retail stores	Retail trade	2.26
Hotels & gaming	Hotels & lodging places	1.57
Textiles	Broad & narrow fabrics, yarn & thread mill	1.42
Drugs	Drugs	1.31
Publishing	Printing & publishing	1.31
Tire & rubber	Tire & inner tubes	1.22
Machine tool	Metalworking machinery & equipment	1.20
Toys & school supplies	Games, toys & children's vehicles	1.16
Electrical equipment	Electrical industrial equipment	1.10
Metal fabricating	Fabricated structural metal	1.09
Toiletries & cosmetics	Toilet preparations	1.08
Health care & hosp. supply	Surgical & medical instruments	1.08
Apparel	Apparel	1.00
Packaging & containers	Glass containers	0.98
Precision instruments	Scientific & controlling instruments	0.95
Original equipment auto parts	Motor vehicle parts & accessories	0.74
Replacement auto parts	Motor vehicle parts & accessories	0.74
Multiform	Miscellaneous machinery	0.72

Table 12: energy dependence

Table 12 (continued)

Value Line industry	Input-output industry	Energy fraction
Mfg. housing & rec. vehicles	Mobile homes	0.72
Medical services	Hospitals	0.71
Computer data processing	Electronic computing equipment	0.69
Office equipment & supplies	Office, computing & accounting machines	0.66
Industrial services	Other business services	0.65
Machinery	General industrial machinery & equipment	0.63
Entertainment	Motion pictures	0.57
Food processing	Food & kindred products	0.50
Aerospace & defense	Aircraft	0.48
Electronics	Electronic components & accessories	0.43
Construction machinery	Construction machinery & equipment	0.43
Fast food & restaurants	Eating & drinking places	0.36
Home appliances	Household appliances	0.25
Shoes	Shoes, except rubber	0.21
Broadcasting	Radio & TV broadcasting	0.11

Table 13

Change in the import penetration ratio

This table reports the interindustry variation of the change in the import penetration ratio between the 1977-81 and 1982-89 periods for the 51 Value Line industries. The import penetration ratio is the ratio of imports divided by new supply. Import avg. 1977-81 is the average of the import penetration ratio for the 1977-81 period. Import avg. 1982-89 is the average of the import penetration ratio for the 1982-89 period. Change is the difference between the import penetration ratios in the 1982-89 and 1977-81 periods. Import data are taken from SIC industries found in various editions of the *U.S. Industrial Outlook* and *U.S. Commodity Exports and Imports as Related to Output*, both compiled by the U.S. Department of Commerce. Value Line and SIC industries are matched using primary and secondary SIC codes.

Value Line industry	SIC industry	Import avg. 1977-81	Import avg. 1982-89	Change
Shoes	Footwear, except rubber	31.18%	55.84%	24.66%
Machine tool	Machine tool, metal cutting	18.04	34.94	16.90
Apparel	Apparel	15.40	29.71	14.31
Construction machinery	Construction machinery	4.90	14.48	9.57
Office equipment & supplies	Office equipment	18.02	26.38	8.35
Computer data processing	Electronic computing	4.70	12.63	7.93
Orig. equip. auto parts	Motor vehicle parts	8.80	14.96	6.16
Replacement auto parts	Motor vehicle parts	8.80	14.96	6.16
Home appliances	Household appliances	8.56	14.11	5.55
Electronics	Electronic components	13.90	19.35	5.45

Table 12 continued and Table 13 start

Table 13 (continued)

Value Line industry	SIC industry	Import avg. 1977-81	Import avg. 1982-89	Change
Tire & rubber	Tire & inner tubes	11.40	16.55	5.15
Machinery	General industrial machinery	4.24	8.80	4.56
Multiform	General industrial machinery	4.24	8.80	4.56
General steel	Steel mill products	11.50	15.80	4.30
Integrated steel	Steel mill products	11.50	15.80	4.30
Electrical equipment	Electrical equipment	5.88	10.00	4.12
Textiles	Textile mill products	4.50	7.28	2.78
Basic chemicals	Chemicals & allied products	4.30	6.53	2.23
Diversified chemicals	Chemicals & allied products	4.30	6.53	2.23
Specialty chemicals	Chemicals & allied products	4.30	6.53	2.23
Precision instruments	Measuring & controlling devices	3.78	5.80	2.02
Toys & school supplies	Games, toys & children's vehicles	19.84	21.80	1.96
Packaging & containers	Glass containers	0.92	2.60	1.68
Drugs	Drugs	4.86	6.50	1.64
Toiletries & cosmetics	Toilet preparations	0.92	2.55	1.63
Aerospace & defense	Aircraft	3.80	5.43	1.63
Integrated petroleum	Petroleum refining	7.40	8.91	1.51
Metal fabricating	Fabricated structural metal	0.68	1.86	1.18
Food processing	Food & kindred products	3.86	4.73	0.87
Paper & paper products	Paper & allied products	7.20	7.95	0.75
Health care & hosp. supply	Surgical & medical instruments	5.60	6.34	0.74
Air transport	Air transportation	0.00	0.00	0.00
Broadcasting	Communication	0.00	0.00	0.00
Building	Building construction	0.00	0.00	0.00
Drugstores	Miscellaneous retail	0.00	0.00	0.00
Entertainment	Motion pictures	0.00	0.00	0.00
Fast food & restaurants	Eating & drinking places	0.00	0.00	0.00
Grocery stores	Food stores	0.00	0.00	0.00
Hotels & gaming	Hotels & lodging places	0.00	0.00	0.00
Industrial services	Business services	0.00	0.00	0.00
Medical services	Health services	0.00	0.00	0.00
Mfg. housing & rec. vehicles	Mobile homes	0.00	0.00	0.00
Natural gas	Electrical, gas & sanitary services	0.00	0.00	0.00
Retail stores	General merchandise	0.00	0.00	0.00
Retail stores, special lines	Miscellaneous retail	0.00	0.00	0.00
Trucking & transport leasing	Motor freight transport & warehouse	0.00	0.00	0.00
Coal, uranium, geothermal	Coal	0.28	0.28	-0.01
Oilfield services	Oil & gas field machinery	1.02	0.98	-0.04
Publishing	Printing & publishing	0.84	0.56	-0.28
Metals & mining	Metal & mining	15.84	12.31	-3.53
Petroleum producing	Crude petroleum & natural gas	41.18	26.91	-14.27

Table 13 continued

6 Short summary

- The paper studies whether 1980s takeover and restructuring activity was driven by industry-level shocks.
- The main sample includes 1,064 Value Line firms from year-end 1981, followed through 1982-1989.
- About 57% of firms experience takeover or restructuring activity during the sample window.
- Activity rates vary sharply across industries.
- Timing of activity also varies sharply across industries, indicating industry-specific waves.
- Broad sales and employment shocks measured in 1977-1981 are positively related to takeover activity in 1982-1989.
- Specific shocks such as deregulation explain important portions of takeover activity.
- R&D intensity is negatively related to takeover activity.
- The paper argues that many takeovers are mechanisms for industries to adjust to fundamental changes.
- This interpretation has implications for takeover waves, spillovers, and post-takeover operating performance.

7 Conclusion

7.1 Core conclusion

Mitchell and Mulherin show that takeover and restructuring activity in the 1980s was strongly related to industry-level shocks. The activity was not randomly distributed across firms or industries and was not merely a uniform market-wide wave. Instead, industries facing major economic changes experienced disproportionately high and temporally clustered takeover and restructuring activity.

7.2 Economic intuition

When industry conditions change, assets may need to be redeployed, capacity may need to be reduced, and ownership/control may need to shift to those best able to operate under the new environment. Takeovers, mergers, leveraged buyouts, hostile bids, recapitalizations, and asset restructurings are different institutional mechanisms for achieving this adjustment.

7.3 Takeaway

The paper reframes the 1980s takeover wave as an industry-shock phenomenon. Corporate control activity should therefore be studied not only as firm-level governance discipline but also as an economy-wide asset reallocation mechanism triggered by deregulation, technological change, input-price shocks, and other fundamental disturbances.